

Bayesian Fit Analysis – Kerr–de Sitter Cosmology with Accretion

1. The rSTF Cosmological Conjecture

The Kerr–de Sitter (KdS) cosmological framework presented here stems from the conjecture that our observable universe is embedded within the rotating space-time fabric (rSTF) of a higher-dimensional black hole structure. Rather than relying on a static cosmological constant Λ , this model introduces an acceleration term A_{acc} that encapsulates late-time effects due to angular momentum transfer and mass accretion onto a parent cosmological black hole.

Key assumptions:

- Space-time curvature is influenced by mass-energy (Ω_m), vacuum energy (Ω_Λ), and accretion-driven acceleration (A_{acc}).
- Observers within the rotating frame will not directly measure rotation but will infer apparent acceleration.
- Late-time cosmic acceleration arises dynamically without requiring a fine-tuned Λ constant.

2. Datasets Used

This study incorporates the following observational datasets:

- Pantheon+SH0ES Type Ia Supernovae: 1048 real data points extracted from the FITS file (ally_shoes_ceph_topantheonwt6.0_112221.fits), including redshift z , distance modulus $\mu(z)$, and observational uncertainties.
- BAO: Baryon Acoustic Oscillation measurements from eBOSS DR16.
- $H(z)$: Direct expansion rate measurements at various redshifts.
- CMB: Compressed Planck 2018 likelihood using distance priors (R and l_a).
- GW and Lensing: Constraints on H_0 and S_8 using Gaussian priors.

3. Key Cosmological Equations

Hubble parameter:

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda + A_{\text{acc}}(1 - \exp(-z/0.5))}$$

Distance modulus:

$$\mu(z) = 5 * \log_{10}[(c/H_0) * (1+z) * \int dz'/E(z')] + 25$$

Angular diameter distance and comoving distance functions are used for BAO and CMB integration calculations.

$$\Omega_{\text{total}} = \Omega_m + \Omega_\Lambda + A_{\text{acc}}$$

4. Graphical Output and Interpretation

5. Python Code Structure Overview

- Data Load: Extracts real Pantheon+SHOES data and other datasets (BAO, $H(z)$, CMB).
- Cosmology Model: Defines $E(z)$, distance modulus, and integrals.
- Likelihood: Combines SN χ^2 , BAO χ^2 , $H(z)$ χ^2 with priors from CMB, GW, and S8.
- MCMC: emcee sampler with walkers=32 and 3000 steps.
- Posterior Analysis: Extracts chain, quantiles, Ω_{total} , and visual diagnostics.

6. Summary and Next Steps

The current run produces an overestimated $\Omega_{\text{total}} \sim 2.45$ and a poorly matching $\mu(z)$ curve, with observed SN Ia data far below the model prediction. This suggests that either the priors, the acceleration formulation A_{acc} , or the fitting process require refinement.

Recommendations:

- Reassess prior bounds to allow better parameter exploration.
- Introduce parameter reparameterization or rescaling to address degeneracies.
- Check for data normalization or unit consistency in input sets.
- Test reduced models (e.g., Λ CDM-only) as a baseline.
- Include corner plots for validation against literature constraints.